



National Textile University

Department of Computer Science

Subject: Operating System

Submitted to: Sir Nasir

Submitted by:Eman Tariq

Reg. number: 23-NTU-CS-1150

Lab 12

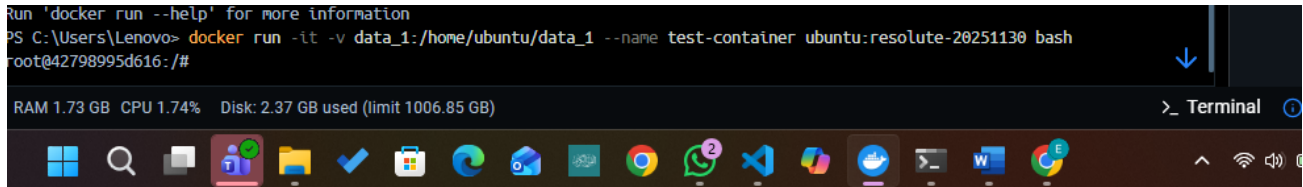
Semester:5th

Data Persistence, Custom Images, Networking & Docker Compose

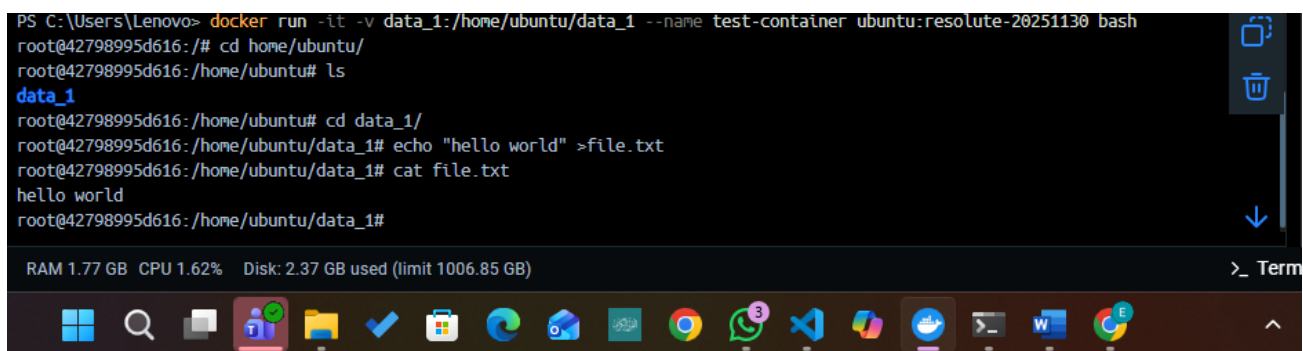
Types of Docker Storage

1. Volumes :

```
Run 'docker run --help' for more information
PS C:\Users\Lenovo> docker run -it -v data_1:/home/ubuntu/data_1 --name test-container ubuntu:resolute-20251130 bash
root@42798995d616:/#
```



```
PS C:\Users\Lenovo> docker run -it -v data_1:/home/ubuntu/data_1 --name test-container ubuntu:resolute-20251130 bash
root@42798995d616:/# cd /home/ubuntu/
root@42798995d616:/home/ubuntu# ls
data_1
root@42798995d616:/home/ubuntu# cd data_1/
root@42798995d616:/home/ubuntu/data_1# echo "hello world" >file.txt
root@42798995d616:/home/ubuntu/data_1# cat file.txt
hello world
root@42798995d616:/home/ubuntu/data_1#
```

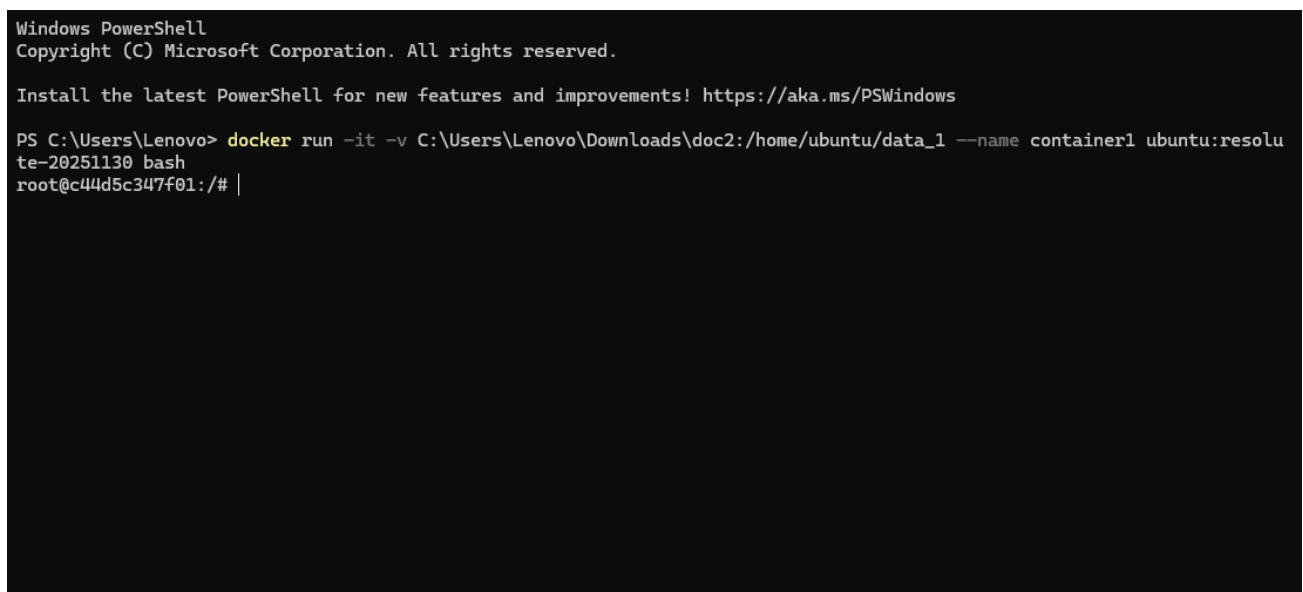


Bind Mount:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Lenovo> docker run -it -v C:\Users\Lenovo\Downloads\doc2:/home/ubuntu/data_1 --name container1 ubuntu:resolute-20251130 bash
root@c44d5c347f01:/#
```



Part 2: Building Custom Images with Dockerfile:

Npm init -y:

```
PS E:\Dockerapp> npm init -y
Wrote to E:\Dockerapp\package.json:

{
  "name": "dockerapp",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

Npm I express:

```
PS E:\Dockerapp> npm i express

added 65 packages, and audited 66 packages in 4s

22 packages are looking for funding
  run `npm fund` for details

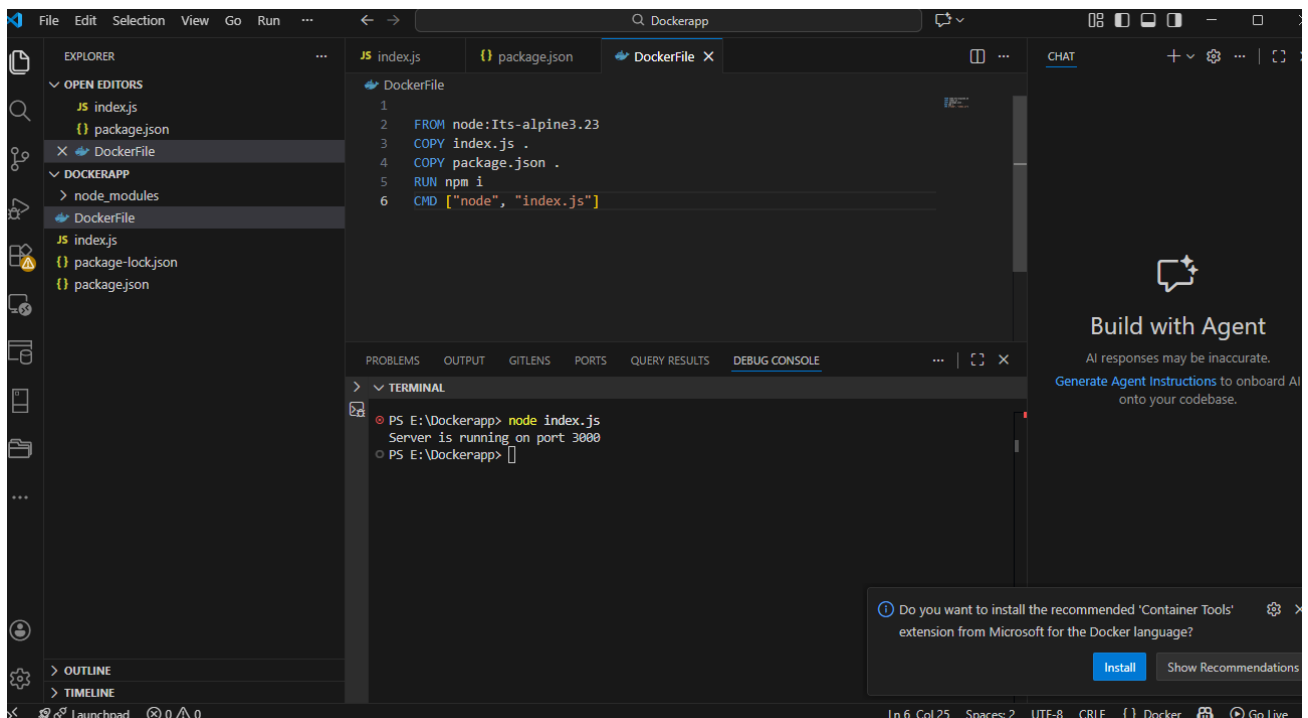
found 0 vulnerabilities
PS E:\Dockerapp> 
```

Ln 5, Col 3 Space

Hello World from Express in Docker!



Commands in Docker File:



Making Image:

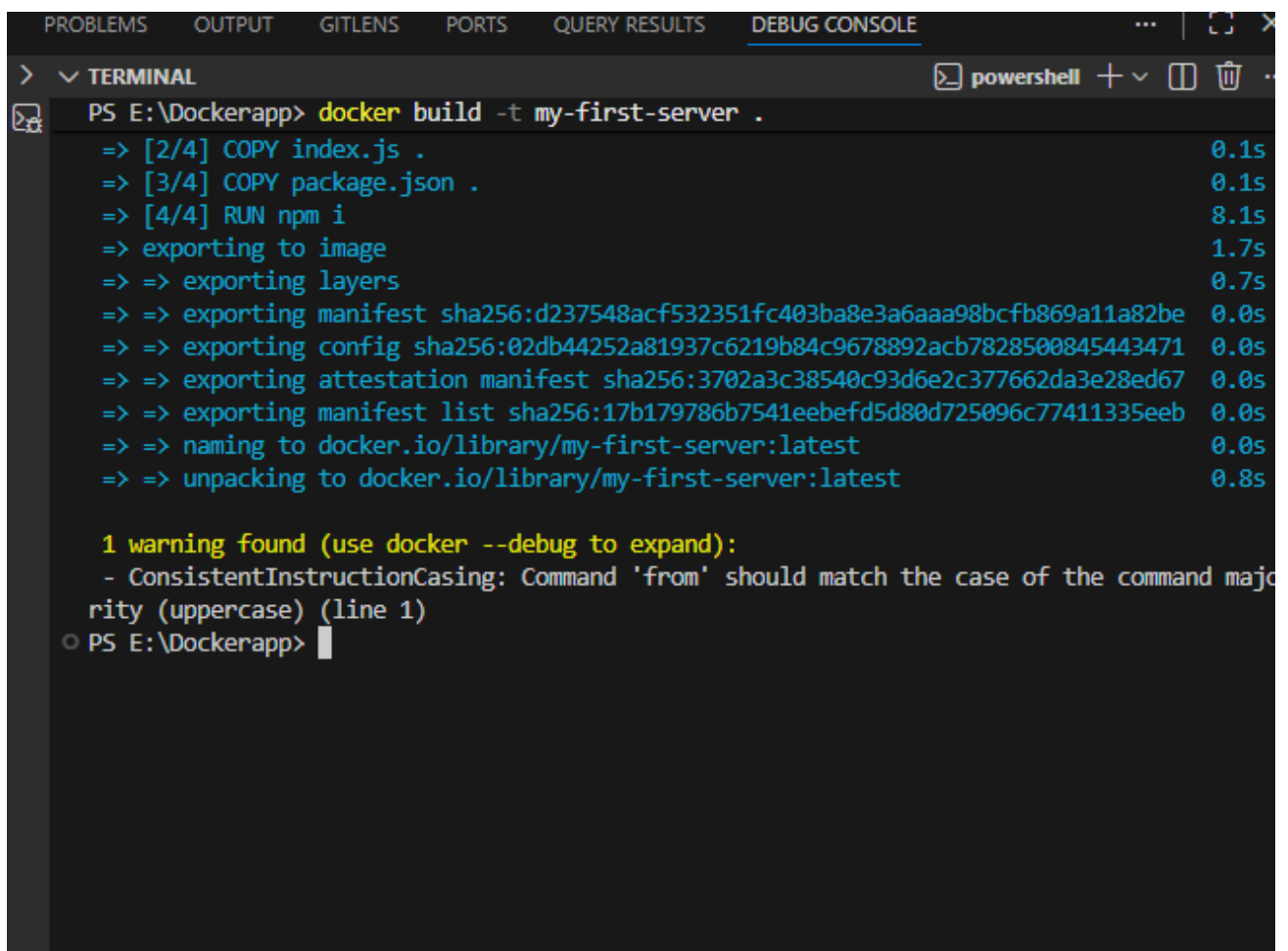
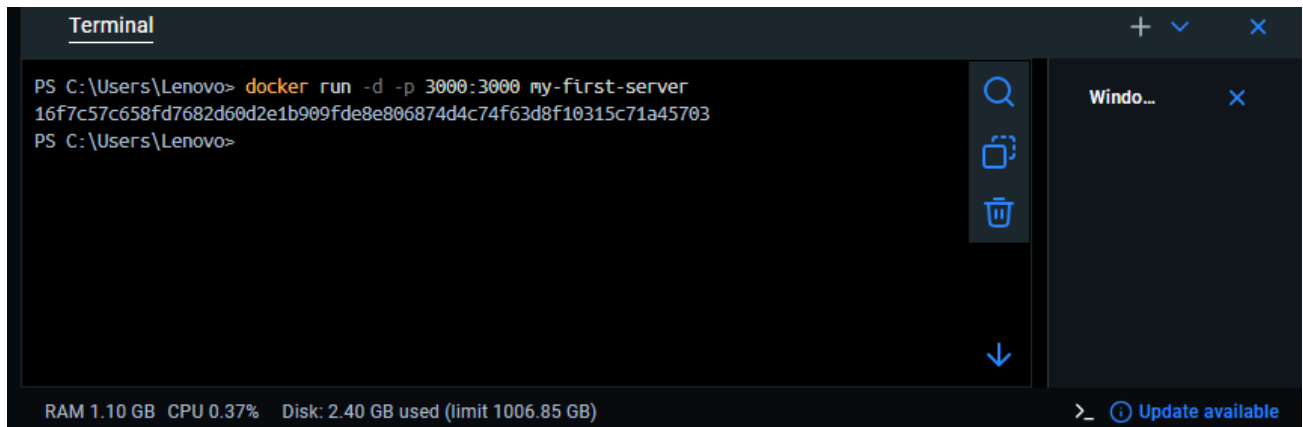


Image Running:



Practice Exercise: Build Your Own Image

Challenge: Serve static HTML files from our previously build node server!

Hello from Docker



JS index.js X {} package.json DockerFile <> index.html {} package-lock.js ...

JS index.js > ...

```
1  import express from 'express'
2
3  const app = express()
4
5  app.use(express.static('public'))
6
7  app.get('/', (req, res) => {
8    |   res.sendFile('index.html', { root: 'public' })
9  | })
10
11 app.listen(6000, '0.0.0.0', () => {
12 |   console.log('Server running on port 6000')
13 | })
14
15
```

VS Code
VS Code
VS Code
VS Code
VS Code

JS index.js

package.json

DockerFile

index.html X

package-lock.js

public > index.html > html > head > title

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>My First Docker Server</title>
5 </head>
6 <body>
7   <h1>Hello from Docker </h1>
8 </body>
9 </html>
```